

- 1 (a) systematic: e.g. constant error (in all readings)
cannot be eliminated by averaging
error in measuring instrument B1
- random: e.g. readings scattered (equally) about true value
error due to observer
can be eliminated by averaging
(*only if averaging not included for systematic*) B1 [2]
- (b) $15 = \pi \times R^2 \times 20$
 $R = 0.4886 \text{ cm}$ (accept any number of s.f.) C1
 % uncertainty in $V = 3.3 \%$ (or $0.5/15$) C1
 % uncertainty in $L = 0.5 \%$ (or $0.1/20$) C1
 % uncertainty in $R = 1.9 \%$ (i.e. one half of the sum) C1
 $R = 0.489 \pm 0.009 \text{ cm}$ A1 [5]